

--	--	--	--	--	--	--	--

B.Tech. Degree VI Semester Regular Examination April 2022

ME 19-205-0604 HEAT AND MASS TRANSFER (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Understand the basic modes of heat transfer.

CO2: Get an insight on conduction, convection and radiation heat transfers.

CO3: Apply the knowledge of heat transfer through various geometries for optimizing the thickness of insulation.

CO4: Understand the unsteady heat conduction and its applications.

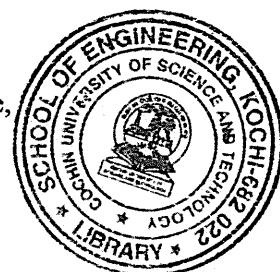
CO5: Get a concept on multiphase flow, diffusion and convective mass transfer.

CO6: Attain information of parallel and counterflow heat exchangers and their design aspects.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze,

L5 – Evaluate, L6 – Create

PO – Programme Outcome



PART A (Answer *ALL* questions)

I.	(8 × 3 = 24)	Marks	BL	CO	PO
(a) Explain unsteady state heat conduction with an example.		3	L2	4	1
(b) Consider two finned surfaces that are identical except that the fins on the first surface are formed by casting whereas they are attached to the second surface afterwards by welding. Identify the case in which the fins will provide greater enhancement in heat transfer. Explain.		3	L3	1	3
(c) State the governing law for Diffusion mass transfer.		3	L1	5	1
(d) Explain development of hydrodynamic and thermal boundary layers with suitable figure.		3	L2	2	1
(e) Define shape factor and state its physical significance.		3	L1	2	1
(f) Explain radiation shields and mention some of their applications.		3	L2	2	1
(g) Show the temperature variation along the length of heat exchanger when hot and cold fluids flow in parallel and counter flow arrangement.		3	L1	6	1
(h) What do you mean by “Fouling” in heat exchangers? How does it affect the performance of heat exchanger?		3	L1	6	1

(P.T.O.)

PART B

(4 × 12 = 48)

- | | | | | | |
|-----------|--|---|----|---|---|
| II. (a) | Explain the significance of critical radius of insulation. | 4 | L2 | 3 | 1 |
| (b) | A steam pipe 75 mm inside diameter and 90 mm outside diameter is lagged with two successive layers of insulation. The layer in contact with the pipe is 38 mm asbestos and the asbestos layer is covered with 25 mm thick magnesia insulation. The surface coefficients for inside and outside surfaces are 227 W/m ² K and 6.8 W/m ² K respectively. If the steam temperature is 375°C and the ambient temperature is 35°C, calculate the steady state loss of heat from steam for 60 m length of pipe. Determine the overall coefficient of inside and outside surfaces of the lagged steam pipe. Thermal conductivity values are:
Pipe material - 45 W/mK
Asbestos - 0.14 W/mK
Magnesia Insulation - 0.07 W/mK | 8 | L4 | 3 | 2 |
| OR | | | | | |
| III. (a) | Derive equations of temperature distribution and heat dissipation for an infinitely long fin. | 7 | L3 | 3 | 1 |
| (b) | A motor body is 360 mm in diameter (outside) and 240 mm long. Its surface temperature should not exceed 55°C when dissipating 340 W. Longitudinal fins of 15 mm thickness and 40 mm height are proposed. The convection coefficient is 40 W/m ² °C. Determine the number of fins required. Atmospheric temperature is 30°C, thermal conductivity = 40 W/m°C. | 5 | L4 | 3 | 2 |
| IV. (a) | Atmospheric air at 25°C flows over a flat plate at a velocity of 3 m/s. If the plate is 1m wide and 80°C, calculate the following at x = 300 mm.
(i) Hydrodynamic boundary layer thickness
(ii) Thermal boundary layer thickness
(iii) Local friction coefficient
(iv) Average friction coefficient
(v) Local heat transfer coefficient
(vi) Average heat transfer coefficient
(vii) Heat transfer. | 7 | L3 | 2 | 2 |
| (b) | An open pan 20 cm in diameter and 8 cm deep contains water at 25°C and is exposed to dry atmospheric air. If the rate of diffusion of water vapour is 8.54 × 10 ⁴ kg/hr, estimate the diffusion coefficient of water in air. | 5 | L3 | 5 | 2 |
| OR | | | | | |
| V. (a) | A condenser is to be designed to condense 225 Kg of steam per hour at a pressure of 0.15 bar. A square array of 400 tubes, each of 6 mm in diameter, is available for the task. If the tube surface temperature is to be maintained at 26°C, make calculations for the length of each tube. | 6 | L3 | 2 | 2 |
| (b) | Explain diffusivity, diffusional resistance and mass transfer coefficient. Write the units for each term. | 6 | L3 | 5 | 1 |

(Continued)

BTS-VI(R)-04.22-0303

VI.	(a)	Two concentric cylinders have inner and outer radius with 5 cm and 10 cm and length 20 cm. Calculate the view factors.	6	L3	2	2
	(b)	Determine the heat lost by radiation per meter length of 80 mm diameter pipe at 300°C, if	6	L4	2	2
		(i) located in a large room with red brick walls at temperature of 27°C				
		(ii) enclosed in a 160 mm diameter red brick conduit at a temperature of 27°C.				
		Emissivities of pipe material and brick 0.79 and 0.93 respectively.				
		OR				
VII.	(a)	Explain the terms absorptivity, reflectivity and transmissivity.	3	L2	2	1
	(b)	Two large plates, one at 800 K and other at 600 K have emissivities 0.5 and 0.8 respectively. A radiation shield having an emissivity 0.1 on one side and emissivity 0.05 on the other side is placed between the plates. Obtain the heat transfer by radiation per square meter with and without the radiation shield.	9	L4	2	2
VIII.	(a)	Derive an equation for the effectiveness (ϵ) of counter flow heat exchanger in terms of NTU and Capacity Ratio (C).	8	L3	6	1
	(b)	A feed water heater uses steam condensing at 120°C to heat water from 30°C to 90°C in a shell and tube heat exchanger at a rate of 5 kg/s. The overall heat transfer coefficient is equal to 2000 W/m ² K. Due to usage deposits have formed on the surfaces resulting in a fouling resistance of 0.0009 Km ² /W. Determine the water exit temperature if flow rates and inlet temperatures are maintained.	4	L3	6	2
		OR				
IX.	(a)	Derive the expression for LMTD of a parallel flow heat exchanger.	4	L3	6	1
	(b)	A counter flow heat exchanger is used to cool 2000 kg/hr of oil ($C_p = 2.5$ kJ/kgK) from 105°C to 30°C by the use of water entering at 15°C. The overall heat transfer coefficient is expected to be 1.5 KW/m ² K. Presume that the exit temperature of the water is not to exceed 80°C. Using NTU method, make calculations for the	8	L3	6	2
		(i) water flow rate				
		(ii) surface area required for heat transfer				
		(iii) effectiveness of heat exchanger.				

Bloom's Taxonomy Levels

L1 – 16.6 %, L2 – 20.8 %, L3 – 45.8 %, L4 – 16.6 %, L5 – 0 % and L6 – 0 %.
